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Vectorizing by piecewise deconstruction of object strokes

Abstract

In implementation of techniques for vectorizing by piecewise deconstruction of object strokes, a computing device implements an image processing system to receive an input to initiate the generation of a boundary of an object in a digital image, such as a raster image. The image processing system detects a set of visually separated but semantically related strokes that represent the object's boundary. Based on the set of visually separated but semantically related strokes, the image processing system forms a combined stroke and generates the boundary of the object as a path based on the combined stroke. The resulting path mimics the visual appearance of the object in the digital image in a vector space.

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Background/Summary

BACKGROUND

(1) Various hardware and software systems are configured to reproduce different types of digital images as vector images via vectorization. A vector image is a type of digital image that is made up of paths (e.g., points, lines, curves, and shapes) that relate to one another via mathematical formulas. Vector images have several advantages over other types of digital images, such as scalability, small file sizes, editability, clarity of strokes, and versatility.

(2) However, conventional vectorization techniques result in various issues, such as extraneous paths, computational inefficiencies, visual inaccuracies, and extraneous power consumption. Techniques and systems are needed to make vectorization of digital images more effective and efficient.

SUMMARY

(3) Techniques and systems for vectorizing based on piecewise deconstruction of object strokes are described. In one example, a computing device receives an input to initiate the generation of a boundary of an object in a digital image. Based on the input, the computing device detects a set of strokes that are visually separated and semantically related from a plurality of strokes that represent the boundary of the object in the digital image.

(4) In some examples, the computing device forms a combined stroke based on the set of detected strokes. The computing device generates the boundary of the object as a path based on the combined stroke. Examples of the path include vector curves, cubic Bezier curves, fills, and so forth. Based on the path, the computing device generates a boundary of the object as mimicking a visual appearance of the object in the digital image. The disclosed techniques and systems enable efficient and accurate vectorization of objects in digital images by generating the boundary of the object as the path by combining the visually separated but semantically related strokes that form the object's boundary.

(5) This Summary introduces a selection of concepts in a simplified form that are further described below in the Detailed Description. As such, this Summary is not intended to identify essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

(2) The detailed description is described with reference to the accompanying figures. Entities represented in the figures are indicative of one or more entities and thus reference is made interchangeably to single or plural forms of the entities in the discussion.

(3) FIG. 1 is an illustration of a digital medium environment in an example implementation that is operable to employ techniques and systems for vectorizing a digital image based on piecewise deconstruction of object strokes, as described herein.

(4) FIG. 2A depicts a system in an example implementation showing operation of a vectorization module for vectorizing a digital image based on piecewise deconstruction of object strokes by generating a boundary of an object as a path based on the piecewise deconstruction of the object strokes.

(5) FIG. 2B depicts a system in an example implementation showing operation of a vectorization

- module for vectorizing a digital image based on piecewise deconstruction of object strokes by rendering a boundary of an object based on the piecewise deconstruction of the object strokes.
- (6) FIG. 3 depicts a system in an example implementation showing operation of a vectorization module for vectorizing a digital image based on piecewise deconstruction of object strokes by generating stroke colors of an object, fills of an object, and a boundary of an object as a path, and rendering vector colors of the object based on the piecewise deconstruction of the object strokes.
- (7) FIG. 4 depicts an example of receiving an input to initiate generation of a boundary of an object in a digital image.
- (8) FIG. 5 depicts an example of detecting a set of strokes that are visually separated and semantically related from a plurality of strokes that form a boundary of an object in a digital image.
- (9) FIG. 6 depicts an example of forming a combined stroke based on a set of strokes that are visually separated and semantically related.
- (10) FIG. 7A depicts an example of determining medial axes of a set of strokes that are visually separated and semantically related.
- (11) FIG. 7B depicts an example of generating describing display of characteristics of a set of strokes that are visually separated and semantically related in relation to medial axes.
- (12) FIG. 8 depicts an example of generating a boundary of an object as a path based on a combined stroke based on a set of strokes that are visually separated and semantically related.
- (13) FIG. 9 depicts an example of rendering a boundary of an object and vector colors of the object as mimicking a visual appearance of the object based on piecewise deconstruction of object strokes of the object.
- (14) FIG. 10 depicts a procedure in an example implementation of vectorizing a digital image based on piecewise deconstruction of object strokes.
- (15) FIG. 11 illustrates an example system including various components of an example device that can be implemented as any type of computing device as described and/or utilized with reference to FIGS. 1-10 to implement embodiments of the techniques described herein.

DETAILED DESCRIPTION

Overview

- (16) Digital images, such as raster images, are convertible to vector images via vectorization. To reap the advantages of vector images, various hardware and software systems utilize conventional vectorization techniques to vectorize digital images via either manual vectorization by a user or batch vectorization techniques.
- (17) Manual vectorization requires a user to meticulously maneuver a cursor to produce a vectorized version of a digital image. For example, to vectorize a raster image of a circle that has an outline of one hundred black dashes, a user meticulously uses a cursor to reproduce the one black hundred dashes as vector strokes in a single path. Accordingly, manual vectorization is a tedious and time-consuming process. Manual vectorization also requires considerable processing power and time to convert a raster image to a vector image.
- (18) Another example of conventional vectorization techniques is batch vectorization. Below is a description of the example circle object discussed above, but in a batch vectorization context instead of a manual vectorization context. Initially, the user provides a computing device with the raster image including the circle. Based on the raster image, the computing device generates a vector image including one hundred paths in which each path includes one stroked vector that corresponds to one black dash of the circle's outline. In this way, batch vectorization fails to create paths as a user would create them in a vector space. As described in the example of the circle object in the manual vectorization context, the user creates a single stored path instead of one hundred paths. In summary, conventional batch vectorization techniques produce extraneous paths because conventional batch vectorization techniques fail to detect and integrate non-solid strokes that are semantically related into a single path. The outcome of this failure is an extraneous number of paths that is far from being directly usable.

(19) Batch vectorization results in significant user input to repair the inaccuracies and inefficiencies caused by the batch conversion process. Further, producing extraneous paths based on the raster image causes computing devices to waste unnecessary processing power and memory resources. For instance, batch-based vectorization systems can utilize a large amount of processing resources to convert edges from the raster image into vector strokes that are fractured into extraneous paths, which are then removed and/or corrected based on the significant repairing process described above. Accordingly, conventional vectorization techniques result in computational inefficiencies, visual inaccuracies, and extraneous power consumption.

(20) Techniques and systems for vectorizing a digital image based on piecewise deconstruction of object strokes are described that overcome these limitations. In various examples, a computing device implements an image processing system to receive an input to initiate generation of a boundary of an object in a digital image in a user interface. As an example, a user uploads a raster image of a circle to the image processing system. The circle's outline is a black dashed line, formed by one hundred dashes that are visually separated and semantically related. Upon receiving the input, the image processing system initiates the generation of the boundary of the object in the digital image.

(21) Based on the input raster image, the image processing system detects a set of strokes, which are the one hundred visually separated and semantically related dashes that represent the boundary of the circle. In various examples, the image processing system detects the set of strokes that are visually separated and semantically related via a deep neural network.

(22) Following the detection, the image processing system forms a combined stroke based on the set of dashes. To do this, the image processing system combines the one hundred dashes, effectively reconstructing a single combined stroke of the dashed outline of the circle.

(23) Finally, the image processing system generates the boundary of the object as a path, utilizing the combined stroke. This newly created path accurately represents the original dashed outline of the colored circle in the raster image. In this way, instead of inefficiently and ineffectively generating the boundary of the object as one hundred paths via conventional vectorization techniques, the image processing system efficiently generates the boundary of the object as a single path by vectoring based on piecewise deconstruction of object strokes. Vectorizing based on piecewise deconstruction of object strokes solves the problem of semantically related and visually separated strokes being fractured across different paths, such as across one hundred different paths. Vectorizing based on piecewise deconstruction of object strokes also result in computational efficiencies and effective power consumption as compared to conventional vectorization techniques.

(24) In the following discussion, an example environment is described that employs the techniques described herein. Example procedures are also described that are performable in the example environment as well as other environments. Consequently, performance of the example procedures is not limited to the example environment and the example environment is not limited to performance of the example procedures.

Example Environment

(25) FIG. 1 is an illustration of a digital medium environment **100** in an example implementation that is operable to employ techniques and systems for vectorizing a digital image based on piecewise deconstruction of object strokes.

(26) The illustrated environment **100** includes a computing device **102** connected to a network **104**. The computing device **102** is configurable as a desktop computer, a laptop computer, a mobile device (e.g., assuming a handheld configuration such as a tablet or mobile phone), and so forth. Thus, the computing device **102** is capable of ranging from a full resource device with substantial memory and processor resources (e.g., personal computers, game consoles) to a low-resource device with limited memory and/or processing resources (e.g., mobile devices). In some examples, the computing device **102** is representative of a plurality of different devices such as multiple

servers utilized to perform operations “over the cloud.”

(27) The computing device **102** includes an image processing system **106** configured to process images. By way of example the image processing system **106** may be configured as an application for image processing (e.g., for creating and/or editing digital content), a platform for image processing, a software for image processing, and so forth.

(28) The computing device **102** includes a storage device **108**. The storage device **108** is illustrated to include digital content **110**, such as digital images, deep neural networks, and so forth. The image processing system **106** includes a vectorization module **112** configured to vectorize (e.g., via generation or rendering) images, such as raster images. In some examples, the vectorization module **112** is configured to convert digital content (e.g., raster images, vector images, or other non-vector formats of digital content) into vector parts (e.g., paths, fills, strokes, shapes, etc.) or vector formats. The vector formats include but are not limited to Scalable Vector Graphics (“SVG”), Adobe Illustrator (“AI”), Encapsulated PostScript (“EPS”), Portable Document Format (“PDF”), Drawing Exchange Format (“DXF”), Windows Metafile (“WMF”), CorelDRAW (“CDR”), and Enhanced Metafile (“EMF”).

(29) The computing device **102** includes a user interface **114** configured to display digital content **110** and to receive various input **116**, such as a digital image **118** as input **116**. Examples of the digital image **118** include vector images, raster images, and so forth. In some examples, the input **116** includes input **116** provided by a user, such as via an input device (e.g., a mouse, a keyboard, a stylus, a microphone, etc.) and the input device communicates the input **116** to the computing device **102**, for example, via the network **104**. The input **116** describes the user's interactions with the input device. In various examples, the user interacts with the input device and the image processing system **106** to initiate vectorization of the digital image **118**. Examples of the initiation of the vectorization include initiation of generation or rendering of a path **122** of the object depicted in the digital image **118**.

(30) The image processing system **106** and the vectorization module **112** are configured to receive the input **116**. The image processing system **106** and the vectorization module **112** are further configured to provide an output **120** based on the input **116**, in which the output includes a path **122**. In some examples, the path **122** represents a variety of shapes, such as lines, curves, polygons, or complex shapes with multiple curves and angles. In some examples, the path **122** is assigned a stroke, fill, and other visual attributes to customize its appearance.

(31) In general, functionality, features, and concepts described in relation to the examples above and below are employed in the context of the example procedures described in this section. Further, functionality, features, and concepts described in relation to different figures and examples in this document are interchangeable among one another and are not limited to implementation in the context of a particular figure or procedure. Moreover, blocks associated with different representative procedures and corresponding figures herein are applicable together and/or combinable in different ways. Thus, individual functionality, features, and concepts described in relation to different example environments, devices, components, figures, and procedures herein are usable in any suitable combinations and are not limited to the particular combinations represented by the enumerated examples in this description.

Vectorizing by Piecewise Deconstruction of Object Strokes

(32) FIG. 2A depicts a system **200A** in an example implementation showing operation of the vectorization module **112** of FIG. 1 in greater detail. The following discussion describes techniques that are implementable utilizing the previously described systems and devices. Aspects of each of the procedures are implemented in hardware, firmware, software, or a combination thereof. The procedures are shown as a set of blocks that specify operations performed and/or caused by one or more devices and are not necessarily limited to the orders shown for performing the operations by the respective blocks. In portions of the following discussion, reference is made to FIGS. 1-11.

(33) In system **200A**, the vectorization module **112** includes a stroke detection module **202**, a stroke

combining module **204**, and a vector generation module **206**. To begin in this example, the vectorization module **112** is configured to receive an input **116** to initiate vectorization of a digital image **118**. In various examples, the input **116** is configured to initiate the generation of a boundary of an object in a digital image **118**. The input **116** may be provided by a user or an external system and can include, for example, selecting a specific object within the digital image **118** or providing coordinates to define the object's position in the image. In some examples, the input **116** is a selection of an element displayed via the user interface **114** to initiate the generation.

(34) Based on the input **116**, the stroke detection module **202** is configured to detect strokes **208** that are visually separated and semantically related from a plurality of strokes that represent the boundary of the object in the digital image **118**. The stroke detection module **202** may employ various techniques, such as edge detection algorithms, pattern recognition, or machine learning techniques, to identify and isolate the set of strokes **208** that are relevant to the object's boundary.

(35) In various examples, the stroke detection module **202** employs a deep neural network to isolate the plurality of strokes that correspond to the object's boundary. The deep neural network is configured to isolate the plurality of strokes that correspond to the object's boundary in various ways, such as by treating raster fill pixels of the digital image **118** as noise and removing the raster fill pixels of the digital image **118**, by processing the digital image **118** to output a greyscale version of the digital image **118** including the isolated plurality of strokes, by enabling uninterrupted passage of gradients to decoder layers, by employing a misclassification loss function, and so forth. In some embodiments, the deep neural network is trained on data to determine semantic differences between fills and strokes, to identify semantically related strokes such as dashed strokes, and so forth.

(36) Additionally, or alternatively, the stroke detection module **202** employs one or more deep neural networks to semantically segment the plurality of strokes. In various examples, the one or more deep neural networks detect and combine the set of strokes **208** that correspond to the object's boundary and are visually separate and semantically related. Various examples of the one or more deep neural networks include but are not limited to U-Net based deep neural networks, autoencoders, deep neural networks trained with misclassification loss techniques, and so forth. The one or more deep neural networks are configured to detect the set of strokes **208** that are visually separate and semantically related via various techniques, such as by detecting strokes that are not visually separate (e.g., solid strokes), by misclassification loss techniques, by utilizing skip connections to decoder layers from counterparts in encoder layers, and so forth.

(37) Based on the set of strokes **208**, the stroke combining module **204** forms a combined stroke **210**. The stroke combining module **204** is configured to utilize various techniques, such as algorithms, machine learning, and so forth to combine the detected set of strokes **208**. This can include, for instance, connecting the visually separated strokes based on their proximity, orientation, or other relevant attributes to create the combined stroke **210** that visually represents the object's boundary. In various examples, the stroke combining module **204** employs a deep neural network to generate the combined stroke **210** that visually represents the object's boundary via various techniques, such as by adding pixels to fill gaps between adjacent markings of strokes of the set of strokes **208**, by connecting strokes of the set of strokes **208**, and so forth.

(38) Based on the combined stroke **210**, the vector generation module **206** generates the boundary of the object as a path **122**. The vector generation module **206** is configured to convert the combined stroke **210** into a vector format, such as a polyline, a Bézier curve, or a series of line segments, to create the path **122**.

(39) By implementing vectorization via piecewise deconstruction of object strokes, the invention enables efficient generation of object boundaries in digital images in which sets of semantically related and visually separate object strokes are represented paths corresponding to the set of semantically related and visually separate object strokes, even in cases where the object's boundary consists of visually separated and semantically related object strokes. This improves the

computational efficiency of vectorization techniques in various image processing systems.

(40) FIG. 2B depicts a system **200B** in an example implementation showing operation of a vectorization module **112** for vectorizing a digital image **118** based on piecewise deconstruction of object strokes by rendering a boundary **214** of an object based on the piecewise deconstruction of the object strokes.

(41) In system **200B**, the vectorization module **112** includes the vector generation module **206** of FIGS. 1 and 2A and a vector rendering module **212**, in which the vector rendering module **212** is configured to render vector outputs, such as vector representations of the digital image **118**, vector parts such as paths, fills, and vector colors, and so forth. The vector generation module **206** is illustrated as including a medial axes module **216** and a stroke profiling module **218**, which facilitate the vector generation of the vector generation module **206**.

(42) To begin this example, the vector generation module **206** initiates vector generation (e.g., path generation) by using the digital content **110**, such as the set of strokes **208**, the combined stroke **210**, and so forth. The medial axes module **216**, is configured to determine medial axes **220** of the set of strokes **208** that are visually separated and semantically related. The medial axes module **216** is configured to determine the medial axes **220** via various techniques, such as computational geometry techniques, for computing the medial axes **220**, which represent the one or more centerlines of the set of strokes **208**. In some examples, a medial axis of the medial axes **220** is a reduced or simplified representation of a stroke of the set of strokes **208**, such as a skeleton (e.g., in an image space) of the stroke or a polyline (e.g., in a vector space) representing the centerline of the stroke. By way of example, use of polylines results in higher computational efficiency because using polylines as medial axes **220** can reduce a number of points required to represent a combined stroke, thus leading to faster processing and less memory usage.

(43) Based on the medial axes **220**, the stroke profiling module **218** generates stroke profiles **222** that describe display characteristics of the set of strokes **208** in relation to the medial axes **220**. These display characteristics may include stroke width, stroke dash length, stroke gap length, stroke color, stroke pattern, stroke opacity, stroke caps, stroke joints, and stroke texture, among others.

(44) The stroke width determines the depicted thickness of a stroke of the set of strokes **208**. Stroke dash length and stroke gap length control the length of the dashes and gaps in the stroke pattern of the stroke of the set of strokes **208**. Stroke color determines the color of the stroke of the set of strokes **208**. Stroke pattern refers to the pattern of dashes, dots, or other markings in the stroke of the set of strokes **208**. Stroke opacity controls the transparency or opacity of the stroke. Stroke caps and stroke joints determine the shape and style of the ends and joins of the stroke. Stroke texture refers to the texture or surface quality of the stroke, such as being rough or smooth. These characteristics can be used in combination to create a wide range of visual effects and styles.

(45) The stroke profiling module **218** is configured to utilize various techniques, such as image processing, pattern recognition, random sampling, or machine learning, to obtain and analyze the display characteristics of the set of strokes **208**. In some embodiments, the stroke profiling module **218** generates stroke profiles **222** by superimposing the combined stroke **210** over the set of strokes **208** and obtaining measurements of the strokes underlying the combined stroke **210**. This superimposing technique provides valuable information on the display characteristics of the set of strokes **208**, such as their widths, dash lengths, or gap lengths. In some examples, the determined measurements for the stroke profiles are modified to a scale proportionate to the digital image **118** of the input **116**, such as the image resolution of the digital image **118** of the input **116**.

(46) In the context of stroke widths, the stroke profiling module **218** is configured to generate stroke profiles **222** by obtaining a plurality of width measurements of a stroke of the set of strokes **208** in relation to a medial axis of the medial axes **220**, identifying the largest width measurement of the stroke in relation to the medial axis, and generating a corresponding stroke profile with the largest width measurement as the stroke width of the stroke profile **222**.

(47) In the context of stroke widths, the stroke profiling module **218** is configured to generate stroke profiles **222**, by obtaining a plurality of dash lengths and a plurality of gap lengths of strokes, when applicable, of the set of strokes **208** in relation to a medial axis of the medial axes **220**. By way of example, a dash length or a gap length would not be applicable to a solid stroke.

(48) The stroke profiling module **218** is configured to identify a dash length by measuring along the medial axis of corresponding strokes, identifying the largest dash length measurement of the stroke in relation to the medial axis, and generating or modifying a corresponding stroke profile with the largest dash length measurement as the dash length of the stroke profile.

(49) The stroke profiling module **218** is configured to identify a gap length by measuring along the medial axis of corresponding strokes, identifying the largest gap length measurement of the stroke in relation to the medial axis, and generating or modifying a corresponding stroke profile with the largest gap length measurement as the gap length of the stroke profile.

(50) In various embodiments, a vector rendering module **212** renders the boundary **214** of the object based on the path **122** and the stroke profiles **222**, mimicking the visual appearance of the object in the digital image **118**. This rendering can create a more accurate representation of the object's boundary, which can be useful for editing, extracting, or incorporating the object into other graphics or designs. By implementing this extended method, the invention provides an enhanced solution for generating object boundaries in digital images, taking into account the display characteristics of the visually separated and semantically related strokes, and offering improved accuracy and flexibility in various digital image processing applications.

(51) Based on the path **122** and the stroke profiles **222**, the vector rendering module **212** renders the boundary **214** of the object as mimicking the visual appearance of the object in the digital image **118**. These rendering techniques result in computationally efficient and accurate representations of the object's boundary in which semantically related and visually separated portions of the boundary are represented in a unified manner in respective paths in a vector space.

(52) FIG. 3 depicts a system **300** in an example implementation showing operation of a vectorization module **112** of FIG. 1 for vectorizing a digital image **118** based on piecewise deconstruction of object strokes by generating stroke colors of an object of a digital image, fills of the object, and a boundary of the object as a path **122**, and rendering vector colors of the object based on the piecewise deconstruction of the object strokes.

(53) The vector generation module **206** of illustrated system **300** includes a vector rendering module **212**. The vector rendering module **212** processes digital content such as paths **122**, stroke colors **304**, and fills **306** to render vector content. By way of example, the vector rendering module **212** is configured to receive the path **122** and the stroke profiles **222** to generate the boundary **214** in the example implementation of system **200B** described above.

(54) The vector rendering module **212** uses the digital content **110** to output rendered vector content that mimics the visual appearance of the object depicted in the digital image **118**. In some examples, the vector rendering module **212** outputs various visual attributes such as vector colors **302** (e.g., stroke colors **304** and colors of fills **306**) and/or the boundary **214** of the object. By way of example, the vector rendering module can output the boundary **214** in which the boundary **214** is colored based on the vector colors **302**. The vector rendering module **212** ensures that the visual appearance of these outputs is consistent with the original digital image **118**, such that the vector colors **302** and the boundary **214** appear as similar as possible to the original colors and edges of the depicted object of the original digital image **118**.

(55) To begin this example, the vector generation module **206** determines colors of the digital image **118** that correspond to the object in the digital image **118**, as well as the locations of these colors. The vector generation module **206** can utilize various color detection and analysis techniques to identify the relevant colors and their locations within the object. In various embodiments, the vector generation module **206** generates the fills **306** and the stroke colors **304** of the object based on the colors of the digital image **118** that correspond to the object and the

locations of these colors. These cross-analysis techniques based on the colors of the digital image **118** that correspond to the object and the locations of these colors can provide a more accurate and visually consistent representation of the object in the digital image **118**.

(56) In some examples, the vector generation module **206** determines colors of the digital image **118** that correspond to a stroke of the set of strokes **208** and locations of these colors that correspond to the stroke of the set of strokes **208**. This information can be used by the vector generation module **206** to generate appropriate fills **306** and stroke colors **304** for the object's boundary. In various embodiments, the system determines whether the colors of the digital image that correspond to the stroke are the same or not. If the colors are not the same, the generation of the fills **306** can be based at least in part on this information, resulting in more accurate fills by the vector generation module **206** for the object's boundary. By way of example, if the colors are not the same, the vector generation module **206** is configured to convert corresponding content of the path **122** (e.g., a vector curve) to one fill of the fills **306** instead.

(57) In some examples, the vector generation module **206** determines whether the colors of the digital image **118** that correspond to the stroke are the same. If the colors are the same, the vector generation module **206** generates the stroke colors **304** based at least in part on this information, and thus generates accurate stroke colors for the object's boundary.

(58) In one or more embodiments, the vector generation module **206** determines fill styles based on the colors of the digital image **118** that correspond to the object and the locations of these colors. The generation of the fills **306** by the vector generation module **206** can be based on these fill styles, which may include solid color fills, linear gradient fills, radial gradient fills, conical gradient fills, diamond gradient fills, pattern fills, image fills, hatching fills, crosshatching fills, texture fills, unknown fills, or noise fills, among others.

(59) Solid color fills apply a single color to a shape. Linear gradient fills create a gradient of colors that blend smoothly between two or more points in a straight line. Radial gradient fills create a gradient of colors that blends smoothly from the center point of the shape outward. Conical gradient fills create a gradient of colors that blend smoothly around a central point, in a conical shape. Diamond gradient fills create a gradient of colors that blend smoothly between four points arranged in a diamond shape. Pattern fills fill the shape with a repeating pattern or texture. Image fills fill the shape with a specific image or picture. Hatching fills fill the shape with parallel lines that create a pattern or texture. Crosshatching fills fill the shape with intersecting lines that create a pattern or texture. Texture fills fill the shape with a specific texture or surface quality, such as wood grain or metal. Noise fills fill the shape with random variations in color or shading, creating a textured or speckled appearance. Each of these example fill styles, among others, can be used by the vectorization module **112** (e.g., the vector generation module **206** or the vector rendering module **212**) to create different visual effects and textures to provide a more accurate representation of the colors of the object depicted in the digital image **118**.

(60) Based on the path **122**, the stroke colors **304**, and the fills **306**, the vector rendering module **212** renders vector colors **302** of the object as mimicking the visual appearance of the object in the digital image **118**. These rendering techniques result in an accurate representation of the object in the digital image **118**, and results in output of the vector colors **302** without the computationally inefficient output of extraneous paths as with the conventional vectorization techniques.

(61) FIGS. **4**, **5**, **6**, **7A**, **7B**, **8**, and **9** illustrate an example of vectorizing by piecewise deconstruction of object strokes. FIG. **4** depicts an example **400** of receiving an input to initiate generation of a boundary of an object in a digital image **402**. As illustrated in example **400**, the vectorization module **112** receives an input to initiate generation of the boundary of an object **404** in the digital image **402**. The digital image **402** includes the object **404** in which the object **404** is a colored circle with a linear gradient fill of purple to pink to orange, in which an outline of the object **404** is a black dashed line of eight visually separate and semantically related black dashes.

(62) FIG. **5** depicts an example **500** of detecting a set of strokes that are visually separated and

semantically related from a plurality of strokes that form a boundary of the object **404** in the digital image **402**. As illustrated in this example **500**, the stroke detection module **202** detects a set of strokes **502** in which the set of strokes **502** includes strokes **502(a)-502(h)** because the strokes **502(a)-502(h)** are detected as being semantically related and visually separated. In some examples, the stroke detection module detects the set of strokes **502** via a deep neural network.

(63) FIG. **6** depicts an example **600** of forming a combined stroke based on a set of strokes that are visually separated and semantically related. As illustrated in this example **600**, the stroke combining module **204** combines the strokes **502(a)-502(h)** of the set of strokes **502** to generate a combined stroke **602**. In various examples, the stroke combining module **204** is configured to generate the combined stroke **602** via a deep neural network. By way of example, the stroke combining module **204** is configured to employ the deep neural network to connect the strokes **502(a)-502(h)** of the set of strokes **502** to generate the combined stroke **602**.

(64) FIG. **7A** depicts an example **700A** of determining medial axes of a set of strokes that are visually separated and semantically related. As illustrated in this example **700A**, the medial axes module **216** of the vector generation module **206** is configured to determine the medial axes **702** of the set of strokes **502**. In various examples, the medial axes **702** are determined via a reduction of the combined stroke **602**. As depicted in this example **700A**, the combined stroke **602** is reduced to the medial axis **702**. In some examples, the medial axis **702** (e.g., a reduced representation of the combined stroke **602**) is converted to a polyline.

(65) FIG. **7B** depicts an example **700B** of generating stroke profiles describing display of characteristics of a set of strokes that are visually separated and semantically related in relation to medial axes **702**. As illustrated in this example **700B**, the stroke profiling module **218** is configured to generate stroke profiles based on strokes **502(a)-502(h)** of the set of strokes **502**. In various examples, the medial axes **702**, such as medial axis **702**, are superimposed over the set of strokes **502** to obtain measurements to generate the stroke profiles **222**. Display characteristics of the stroke profiles **222** may include gap length **704**, dash length **706**, or stroke width **708**, among other display characteristics. In various examples, the stroke profiling module **218** utilizes sampling techniques to obtain accurate measurements to generate the stroke profiles **222**.

(66) FIG. **8** depicts an example **800** of generating a boundary of an object **404** as a path **804** based on a combined stroke based on a set of strokes that are visually separated and semantically related. As illustrated in this example **800**, the vector generation module **206** generates a boundary of the object **404** as a path **804** (including dashes **804(a)-804(h)**) as an output **802** based on the combined stroke **602** of FIG. **6** and the set of strokes **502** (including strokes **502(a)-502(h)**) that are visually separated and semantically related. In this example **800**, the path **804** is illustrated as including vector curves and as being a closed path. In some examples, the path **804** is generated as an open path.

(67) FIG. **9** depicts an example **900** of rendering a boundary **904** of an object **404** and vector colors of the object **404** as mimicking a visual appearance of the object **404** based on piecewise deconstruction of object strokes of the object **404**. As illustrated in this example **900**, the vector rendering module **212** renders the boundary **904** (including dashes **904(a)-904(h)**) of the object **404** and in addition to vector colors (e.g., the vector fill of the shape **908** with a gradient fill style and the black stroke colors of the boundary **904**) as an output **902**, based on the piecewise deconstruction of the object strokes of the object **404** depicted in the digital image **402**.

(68) In general, functionality, features, and concepts described in relation to the examples above and below are employed in the context of the example procedures described in this section. Further, functionality, features, and concepts described in relation to different figures and examples in this document are interchangeable among one another and are not limited to implementation in the context of a particular figure or procedure. Moreover, blocks associated with different representative procedures and corresponding figures herein are applicable individually, together, and/or combined in different ways. Thus, individual functionality, features, and concepts described

in relation to different example environments, devices, components, figures, and procedures herein are usable in any suitable combinations and are not limited to the particular combinations represented by the enumerated examples in this description.

Example Procedures

(69) The following discussion describes techniques which are implementable utilizing the previously described systems and devices. Aspects of the procedure are implementable in hardware, firmware, software, or a combination thereof. The procedure is shown as a set of blocks that specify operations performed by one or more devices and are not necessarily limited to the orders shown for performing the operations by the respective blocks. In portions of the following discussion, reference is made to FIGS. **1-9**.

(70) FIG. **10** depicts a procedure **1000** in an example implementation of vectoring by piecewise deconstruction of object strokes. An input to initiate generation of a boundary of an object in a digital image is received (block **1002**). For instance, in some examples, the vectorization module **112** of an image processing system **106** receives a digital image, such as a raster image, to initiate the generation. Responsive to the input, a set of strokes that are visually separated and semantically related from a plurality of strokes that represent the boundary of the object in the digital image are detected (block **1004**). For example, a deep neural network of the stroke detection module **202** is employable to detect a set of strokes **208** that are visually separated and semantically related, such as a set of dashed strokes forming a shape. A combined stroke is formed based on the set of strokes (block **1006**). For example, the stroke combining module **204** of the vectorization module **112** generates a combined stroke **210** based on the set of strokes **208** received from the stroke detection module **202**. The boundary of the object as a path is generated based on the combined stroke (block **1008**). By way of example, the vector generation module **206** of the vectorization module **112** generates a path **122** based at least in part on the combined stroke **210** received from the stroke combining module **204**. In some embodiments, the path **122** includes vector curves, such as Bezier curves.

Example System and Device

(71) FIG. **11** illustrates an example system generally at **1100** that includes an example computing device **1102** that is representative of one or more computing systems and/or devices that implement the various techniques described herein. This is illustrated through inclusion of the vectorization module **112**. The computing device **1102** is configurable, for example, as a server of a service provider, a device associated with a client (e.g., a client device), an on-chip system, and/or any other suitable computing device or computing system.

(72) The example computing device **1102** as illustrated includes a processing system **1104**, one or more computer-readable media **1106**, and one or more I/O interface **1108** that are communicatively coupled, one to another. Although not shown, the computing device **1102** further includes a system bus or other data and command transfer system that couples the various components, one to another. A system bus includes any one or combination of different bus structures, such as a memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or local bus that utilizes any of a variety of bus architectures. A variety of other examples are also contemplated, such as control and data lines.

(73) The processing system **1104** is representative of functionality to perform one or more operations using hardware. Accordingly, the processing system **1104** is illustrated as including hardware element **1110** that is configurable as processors, functional blocks, and so forth. This includes implementation in hardware as an application specific integrated circuit or other logic device formed using one or more semiconductors. The hardware elements **1110** are not limited by the materials from which they are formed or the processing mechanisms employed therein. For example, processors are configurable as semiconductor(s) and/or transistors (e.g., electronic integrated circuits (ICs)). In such a context, processor-executable instructions are electronically-executable instructions.

(74) The computer-readable storage media **1106** is illustrated as including memory/storage **1112**. The memory/storage **1112** represents memory/storage capacity associated with one or more computer-readable media. The memory/storage **1112** includes volatile media (such as random access memory (RAM)) and/or nonvolatile media (such as read only memory (ROM), Flash memory, optical disks, magnetic disks, and so forth). The memory/storage **1112** includes fixed media (e.g., RAM, ROM, a fixed hard drive, and so on) as well as removable media (e.g., Flash memory, a removable hard drive, an optical disc, and so forth). The computer-readable media **1106** is configurable in a variety of other ways as further described below.

(75) Input/output interface(s) **1108** are representative of functionality to allow a user to enter commands and information to computing device **1102**, and also allow information to be presented to the user and/or other components or devices using various input/output devices. Examples of input devices include a keyboard, a cursor control device (e.g., a mouse), a microphone, a scanner, touch functionality (e.g., capacitive or other sensors that are configured to detect physical touch), a camera (e.g., employing visible or non-visible wavelengths such as infrared frequencies to recognize movement as gestures that do not involve touch), and so forth. Examples of output devices include a display device (e.g., a monitor or projector), speakers, a printer, a network card, tactile-response device, and so forth. Thus, the computing device **1102** is configurable in a variety of ways as further described below to support user interaction.

(76) Various techniques are described herein in the general context of software, hardware elements, or program modules. Generally, such modules include routines, programs, objects, elements, components, data structures, and so forth that perform particular tasks or implement particular abstract data types. The terms “module,” “functionality,” and “component” as used herein generally represent software, firmware, hardware, or a combination thereof. The features of the techniques described herein are platform-independent, meaning that the techniques are configurable on a variety of commercial computing platforms having a variety of processors.

(77) An implementation of the described modules and techniques is stored on or transmitted across some form of computer-readable media. The computer-readable media includes a variety of media that is accessed by the computing device **1102**. By way of example, and not limitation, computer-readable media includes “computer-readable storage media” and “computer-readable signal media.”

(78) “Computer-readable storage media” refers to media and/or devices that enable persistent and/or non-transitory storage of information in contrast to mere signal transmission, carrier waves, or signals per se. Thus, computer-readable storage media refers to non-signal bearing media. The computer-readable storage media includes hardware such as volatile and non-volatile, removable and non-removable media and/or storage devices implemented in a method or technology suitable for storage of information such as computer readable instructions, data structures, program modules, logic elements/circuits, or other data. Examples of computer-readable storage media include but are not limited to RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, hard disks, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other storage device, tangible media, or article of manufacture suitable to store the desired information and are accessible by a computer.

(79) “Computer-readable signal media” refers to a signal-bearing medium that is configured to transmit instructions to the hardware of the computing device **1102**, such as via a network. Signal media typically embodies computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as carrier waves, data signals, or other transport mechanism. Signal media also include any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media include wired media such as a wired network or direct-wired connection, and wireless media such

as acoustic, RF, infrared, and other wireless media.

(80) As previously described, hardware elements **1110** and computer-readable media **1106** are representative of modules, programmable device logic and/or fixed device logic implemented in a hardware form that are employed in some embodiments to implement at least some aspects of the techniques described herein, such as to perform one or more instructions. Hardware includes components of an integrated circuit or on-chip system, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and other implementations in silicon or other hardware. In this context, hardware operates as a processing device that performs program tasks defined by instructions and/or logic embodied by the hardware as well as a hardware utilized to store instructions for execution, e.g., the computer-readable storage media described previously.

(81) Combinations of the foregoing are also be employed to implement various techniques described herein. Accordingly, software, hardware, or executable modules are implemented as one or more instructions and/or logic embodied on some form of computer-readable storage media and/or by one or more hardware elements **1110**. The computing device **1102** is configured to implement particular instructions and/or functions corresponding to the software and/or hardware modules. Accordingly, implementation of a module that is executable by the computing device **1102** as software is achieved at least partially in hardware, e.g., through use of computer-readable storage media and/or hardware elements **1110** of the processing system **1104**. The instructions and/or functions are executable/operable by one or more articles of manufacture (for example, one or more computing devices and/or processing systems **1104**) to implement techniques, modules, and examples described herein.

(82) The techniques described herein are supported by various configurations of the computing device **1102** and are not limited to the specific examples of the techniques described herein. This functionality is also implementable through use of a distributed system, such as over a “cloud” **1114** via a platform **1116** as described below.

(83) The cloud **1114** includes and/or is representative of a platform **1116** for resources **1118**. The platform **1116** abstracts underlying functionality of hardware (e.g., servers) and software resources of the cloud **1114**. The resources **1118** include applications and/or data that can be utilized while computer processing is executed on servers that are remote from the computing device **1102**. Resources **1118** can also include services provided over the Internet and/or through a subscriber network, such as a cellular or Wi-Fi network.

(84) The platform **1116** abstracts resources and functions to connect the computing device **1102** with other computing devices. The platform **1116** also serves to abstract scaling of resources to provide a corresponding level of scale to encountered demand for the resources **1118** that are implemented via the platform **1116**. Accordingly, in an interconnected device embodiment, implementation of functionality described herein is distributable throughout the system **1100**. For example, the functionality is implementable in part on the computing device **1102** as well as via the platform **1116** that abstracts the functionality of the cloud **1114**.

Claims

1. A method comprising: receiving, by a processing device, an input to initiate generation of a boundary of an object in a digital image; responsive to the input, detecting, by the processing device, a set of strokes that are visually separated and semantically related from a plurality of strokes that represent the boundary of the object in the digital image; forming, by the processing device, a combined stroke based on the set of strokes; determining, by the processing device, medial axes of the set of strokes; generating, by the processing device, stroke profiles describing display characteristics of the set of strokes in relation to the medial axes; generating, by the processing device, the boundary of the object as a path based on the combined stroke; and

rendering, by the processing device, the boundary of the object based on the path and the stroke profiles as mimicking a visual appearance of the object in the digital image.

2. The method of claim 1, further comprising generating, by the processing device, the plurality of strokes that represent the boundary of the object in the digital image.

3. The method of claim 2, wherein the generating of the plurality of strokes that represent the boundary of the object in the digital image is via a deep neural network.

4. The method of claim 1, wherein the forming of the combined stroke further comprises connecting strokes of the set of strokes that are visually separated and semantically related.

5. The method of claim 4, wherein the connecting of the strokes of the set of strokes that are visually separated and semantically related includes adding pixels to fill gaps between adjacent markings of the strokes that are visually separated and semantically related.

6. The method of claim 1, wherein the detecting of the set of strokes and the forming of the combined stroke is via a deep neural network.

7. The method of claim 1 wherein: the determining of the medial axes further comprises reducing the combined stroke to the medial axes; and the generating of the stroke profiles further comprises superimposing the combined stroke over the set of strokes and obtaining measurements of the strokes of the set of strokes underlying the combined stroke.

8. The method of claim 1 wherein the generating of the stroke profiles further comprises: obtaining a plurality of width measurements of a stroke of the set of strokes in relation to a medial axis of the medial axes; identifying a largest width measurement of the stroke of the set of strokes in relation to the medial axis; and generating a stroke profile corresponding to the stroke of the set of strokes in which the largest width measurement is a stroke width of the stroke profile.

9. The method of claim 1 wherein the display characteristics include at least one of a stroke width, stroke dash length, or a stroke gap length.

10. The method of claim 1, further comprising: determining, by the processing device, colors of the digital image that correspond to the object in the digital image and locations of the colors of the digital image that correspond to the object in the digital image; and generating, by the processing device, fills and stroke colors of the object based on the colors of the digital image that correspond to the object in the digital image and the locations of the colors of the digital image that correspond to the object in the digital image.

11. The method of claim 10, wherein the determining of the colors and the locations of the colors further comprises determining, by the processing device, colors of the digital image that correspond to a stroke of the set of strokes and locations of the colors of the digital image that correspond to the stroke of the set of strokes.

12. A system comprising: a memory component; and a processing device coupled to the memory component, the processing device to perform operations comprising: receiving an input to initiate generation of a boundary of an object in a digital image; responsive to the input, detecting a set of strokes that are visually separated and semantically related from a plurality of strokes that form the boundary of the object in the digital image; forming a combined stroke based on the set of strokes; determining medial axes of the set of strokes that are visually separated and semantically related; generating stroke profiles describing display characteristics of the set of strokes in relation to the medial axes; generating the boundary of the object as a path based on the combined stroke; and rendering the boundary of the object based on the path and the stroke profiles as mimicking a visual appearance of the object in the digital image.

13. The system of claim 12, further comprising: determining colors of the digital image that correspond to the object in the digital image and locations of the colors of the digital image that correspond to the object in the digital image; and generating fills and stroke colors of the object based on the colors of the digital image that correspond to the object in the digital image and the locations of the colors of the digital image that correspond to the object in the digital image.

14. The system of claim 13, wherein the determining of the colors and the locations of the colors

further comprises determining colors of the digital image that correspond to a stroke of the set of strokes and locations of the colors of the digital image that correspond to the stroke of the set of strokes.

15. The system of claim 14, further comprising determining the colors of the digital image that correspond to the stroke are not same, and wherein the generating of the fills is based at least in part on the colors of the digital image that correspond to the stroke are not the same.

16. The system of claim 14, further comprising determining the colors of the digital image that correspond to the stroke are same colors, and wherein the generating of the stroke colors is based at least in part on the colors of the digital image that correspond to the stroke are the same colors.

17. The system of claim 14, further comprising determining fill styles based on the colors of the digital image that correspond to the object in the digital image and the locations of the colors of the digital image that correspond to the object in the digital image, and wherein the generating of the fills is based on the fill styles.

18. The system of claim 17, wherein the fill styles include at least one of a solid color fill, a linear gradient fill, a radial gradient fill, a conical gradient fill, a diamond gradient fill, a pattern fill, an image fill, a hatching fill, a crosshatching fill, a texture fill, or a noise fill.

19. The system of claim 13, further comprising: based on the path, the stroke colors, and the fills, rendering vector colors of the object as mimicking a visual appearance of the object in the digital image.

20. A non-transitory computer-readable storage medium storing executable instructions, which when executed by a processing device, cause the processing device to perform operations comprising: receiving an input to initiate generation of a boundary of an object in a digital image; responsive to the input, detecting a set of strokes that are visually separated and semantically related from a plurality of strokes that form the boundary of the object in the digital image; determining medial axes of the set of strokes that are visually separated and semantically related; generating stroke profiles describing display characteristics of the set of strokes in relation to the medial axes; forming a combined stroke based on the set of strokes; generating the boundary of the object as a path based on the combined stroke; and rendering the boundary of the object based on the path and the stroke profiles as mimicking a visual appearance of the object in the digital image.
